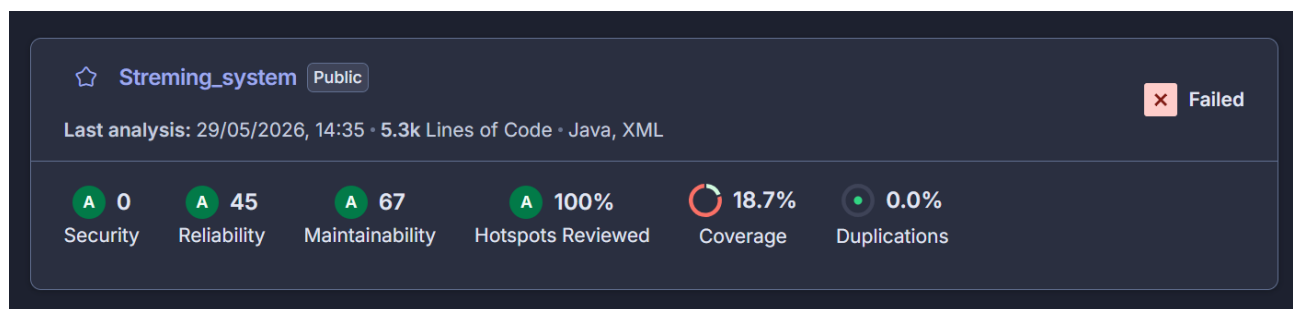


Static testing tool (SonarCloud)

A code quality analysis conducted using SonarQube revealed instances of code duplication within the project. Given that the development team consisted of five members working concurrently, redundant implementations of an email validation function were independently produced by two contributors. The duplication was identified and subsequently addressed through the static analysis capabilities provided by SonarQube.

Beyond duplication detection, SonarQube also served as a valuable tool for security risk evaluation. As the development team lacked substantial academic and practical experience with production-level security standards, the static analysis provided by SonarQube enabled the identification of potential security vulnerabilities that might otherwise have gone undetected. One such vulnerability was discovered in the password change flow, where upon updating a user's password, the previously issued authentication token remained valid for up to one hour. This represents a significant security risk, as it leaves a window of opportunity for unauthorized access in the event that the token has been compromised.

We have failed conditions in SonarCloud because our code coverage is low.



Code coverage report (JaCoCo)

The coverage report showed which packages and methods had low or zero coverage, which highlighted where unit tests were missing. This guided test design by making it clear which classes needed to be tested first. Once tests were written and the report was re-run, changes in coverage percentages confirmed whether the new tests were actually exercising the intended code paths. In this way, the report provided measurable feedback that shaped which tests to write and whether they were sufficient.

Streming_system

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
org.example.mongo.embedded		0 %		0 %	817	817	151	151	421	421	32	32
org.example.entities		13 %		0 %	627	793	88	260	159	325	0	17
org.example.neo4j.nodes		0 %		0 %	663	663	147	147	272	272	17	17
org.example.dtos		23 %		4 %	663	875	13	309	143	354	0	21
org.example.mongo.documents		0 %		0 %	324	324	57	57	161	161	10	10
org.example.neo4j.service		0 %		0 %	129	129	346	346	54	54	1	1
org.example.controllers		46 %		24 %	193	281	232	525	98	186	0	17
org.example.mongo.service		0 %		0 %	93	93	251	251	47	47	1	1
org.example.security.DTOS		16 %		0 %	91	120	4	26	23	52	0	4
org.example.services		94 %		92 %	10	88	6	147	1	31	0	7
org.example.security.services		91 %		77 %	6	21	3	67	2	12	0	2
org.example.security.jwt		90 %		57 %	6	20	2	49	0	13	0	2
org.example		37 %		n/a	1	2	2	3	1	2	0	1
org.example.security		100 %		n/a	0	9	0	26	0	9	0	1
org.example.enums		100 %		n/a	0	6	0	12	0	6	0	6
org.example.security.controllers		100 %		n/a	0	4	0	5	0	4	0	1
org.example.exceptions		100 %		n/a	0	2	0	2	0	2	0	1
Total	26.250 of 31.165	15 %	4.370 of 4.591	4 %	3.623	4.247	1.302	2.383	1.382	1.951	61	141